MPRF: Interpretable Stance Detection through Multi-Path Reasoning Framework

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Abstract

Stance detection, a critical task in Natural Language Processing (NLP), aims to identify the attitude expressed in text toward specific targets. Despite advancements in Large Language Models (LLMs), challenges such as limited interpretability and handling nuanced content persist. To address these issues, we propose the **Multi-Path Reasoning Framework (MPRF)**, a novel framework that generates, evaluates, and integrates multiple reasoning paths to improve accuracy, robustness, and transparency in stance detection. Unlike prior work that relies on single-path reasoning or static explanations, MPRF introduces a structured end-to-end pipeline: it first generates diverse reasoning paths through predefined perspectives, then dynamically evaluates and optimizes each path using LLM-based scoring, and finally fuses the results via weighted aggregation to produce interpretable and reliable predictions. Extensive experiments on the SEM16, VAST, and PStance datasets demonstrate that MPRF outperforms existing models. Ablation studies further validate the critical role of MPRF's components, highlighting its effectiveness in enhancing interpretability and handling complex stance detection tasks.

1 Introduction

Stance detection (Hasan and Ng, 2014; Küçük and Can, 2020), the task of determining the attitude expressed in a text towards a specific target, plays a crucial role in applications such as opinion mining (Graells-Garrido et al., 2020), combating misinformation (Lai et al., 2020), and understanding public sentiment (Lei et al., 2024). By analyzing structural and linguistic patterns in stance reasoning, researchers can gain insights into opinion dynamics, address the evolution of harmful behaviors, and foster a more ethical online environment (De Vinco et al., 2024; Graells-Garrido and Baeza-Yates, 2022; Zhang et al., 2023b).

Existing stance detection methods often treat the task as a classification problem, where models output a stance label without providing interpretable reasoning paths (Allaway et al., 2021; De Vinco et al., 2024; Graells-Garrido and Baeza-Yates, 2022; Li and Zhang, 2024; Xu et al., 2022; Yang and Urbani, 2021; Zhang et al., 2023b). This lack of transparency is particularly problematic in complex tasks that involve subtle or ambiguous opinions, such as those expressed in social media content(Gatto et al., 2023).

Recent advances in large language models (LLMs), particularly those leveraging Chain-of-Thought (CoT) prompting (Wei et al., 2022), have demonstrated remarkable reasoning capabilities in tasks such as multihop question answering (Lu et al., 2022) and mathematical problem solving (Wei et al., 2022). These models achieve zero-shot and few-shot success across diverse tasks by utilizing machine-generated instruction-following data and reasoning mechanisms. While prior work has shown that explanations from models like GPT can improve interpretability in stance detection (Zhang et al., 2024a; Taranukhin et al., 2024), they often rely on single-path or heuristic-based reasoning, which may fail to capture diverse perspectives or provide robust explanations.

In this work, we introduce the Multi-Path Reasoning Framework (MPRF), a novel framework designed to address the interpretability challenges in stance detection by generating, evaluating, and integrating multiple reasoning paths. Unlike previous approaches (Ding et al., 2024a; Taranukhin et al., 2024), which often rely on static reasoning or single-path explanations, MPRF introduces a structured and dynamic pipeline: it first generates diverse reasoning paths through predefined perspectives (e.g., separating sentiment from factual analysis), then evaluates each path using LLM-based

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scoring (relevance, evidence strength, logical consistency), and finally fuses the results via weighted aggregation to produce interpretable and reliable predictions. This systematic integration of generation, evaluation, optimization, and fusion represents a meaningful system-level innovation beyond prompt engineering or isolated use of LLM capabilities.

Our contributions are as follows:

- We propose the Multi-Path Reasoning Framework (MPRF), a novel end-to-end framework for stance detection that goes beyond traditional prompt engineering by combining multiple reasoning paths in a structured, iterative, and interpretable manner.

- We design a dynamic evaluation and optimization mechanism that ensures high-quality reasoning paths through LLM-based scoring (relevance, evidence strength, logical consistency) and refinement of weak paths. We introduce a weighted fusion strategy that prioritizes well-supported reasoning paths, improving prediction reliability and interpretability.

- Extensive experiments show that MPRF achieves state-of-the-art performance across multiple datasets in both zero-shot and few-shot settings, while providing interpretable reasoning chains for each prediction. Our detailed ablation studies highlight the importance of reasoning path evaluation and fusion in improving the accuracy and interpretability of stance detection predictions.

2 Related Work

Early stance detection approaches primarily treated the task as a classification problem, relying on traditional machine learning models with handcrafted features (Aldayel and Magdy, 2019; Dey et al., 2017). With the emergence of pretrained language models (PLMs) such as BERT (Devlin et al., 2019; Nguyen et al., 2020), these methods achieved significant improvements by learning features from in-domain or cross-domain datasets (Augenstein et al., 2016; Zhang et al., 2019; Allaway et al., 2021; Liu et al., 2021; Liang et al., 2022b). However, these models often lacked interpretability, as they treated stance detection as a black-box process without explicitly modeling the reasoning steps.

Recent advancements in large language models (LLMs) have significantly enhanced stance detection, with a focus on improving reasoning capabilities through Chain-of-Thought (CoT) prompting (Yao et al., 2024). CoT prompting enables LLMs to perform step-by-step reasoning, achieving state-of-the-art results in complex tasks such as arithmetic, logical reasoning, and stance detection. For example, Wei et al. (2022) and Zhou et al. (2022) demonstrated CoT prompting's effectiveness in multihop reasoning tasks. Ding et al. (2024a) applied multi-step CoT prompting to capture the positional perspective of targets in stance detection, while Fei et al. (2023) proposed decomposing tasks into multiple stages for better predictions. Similarly, Ling et al. (2023) introduced deductive reasoning and iterative verification to enhance task inference. Hardalov et al. (2022) designed a prompt-based framework for cross-language stance detection, Zhu et al. (2024) incorporated soft knowledge during cue fine-tuning to improve context understanding, and Huang et al. (2023) expanded the verbalizer in prompt-tuning with external semantic knowledge. While these methods demonstrate the power of CoT prompting in LLMs, they often rely on static reasoning processes and struggle to adapt to tasks requiring diverse perspectives.

Building on these advancements, we propose the Multi-Path Reasoning Framework (MPRF), a novel framework that addresses the limitations of existing CoT-based methods by generating, evaluating, and integrating multiple reasoning paths.

3 Methodology

In this study, we propose a novel framework for stance detection, called Multi-Path Reasoning Framework (MPRF), which is designed to address the critical challenges in stance detection tasks, particularly the lack of transparency and interpretability. MPRF enhances the accuracy, robustness, and interpretability of stance detection by generating multiple reasoning paths, evaluating and optimizing these paths, and finally combining the results through a weighted fusion mechanism. Our approach is structured into five key steps: generating multiple reasoning paths (3.2), evaluating the paths (3.3), optimizing the paths (3.4), weighted fusion of the paths (3.5), and outputting the stance label along with the interpretable reasoning paths (3.6). MPRF shown in figure 1 ensures that the detected stance is not only accurate but also transparent, providing clear, traceable reasoning behind the final decision.

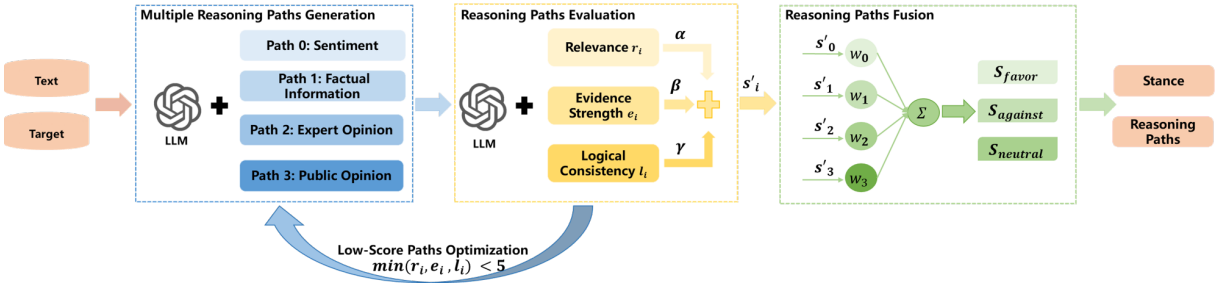


Figure 1: This is the Multi-Path Reasoning Framework. We generate four reasoning paths. A final score s'_i is assigned to each path i by combining the relevance r_i, evidence strength e_i , and logical consistency scores l_i weighted by α, β, and γ after optimization. S_t is the aggregated weighted score for label t, $t \in \{\text{favor}, \text{against}, \text{neutral}\}$. The label with the highest score is selected as the final stance.

3.1 Task Definition

Let $\mathcal{D} = \{(x_j, p_j, y_j)\}_{j=1}^{N}$ be a dataset consisting of N instances, where each instance (x_j, p_j, y_j) represents the input text x_j for which the stance needs to be detected, the corresponding target p_j towards which the stance of x_j is to be determined, and the stance label y_j for the input x_j towards the target p_j, where $y_j \in \{\text{favor}, \text{against}, \text{neutral}\}$.

Stance detection aims to predict the stance label y_j for each input sentence x_j towards the given target p_j.

3.2 Generating Multiple Reasoning Paths

In this step, we generate multiple reasoning paths for stance detection, each representing a distinct chain of thought from a theoretically grounded perspective. This multi-path approach is designed to deconstruct the complex process of stance formation into its fundamental cognitive and social components, ensuring a more robust, accurate, and interpretable analysis. The specific prompts used to guide each reasoning path are detailed in Appendix B.

The first reasoning path is **Sentiment Analysis**. This path is rooted in the understanding that stance is often an expression of affective state. Emotions are powerful drivers of human judgment and decision-making, frequently serving as the primary motivator for taking a position on an issue (Dey et al., 2017). The model identifies key emotional lexicons in the text (e.g., "happy", "angry", "disappointed") to determine the overall valence - positive, negative, or neutral.

The second reasoning path is **Factual Reasoning**. This path addresses the cognitive process of evidence-based justification. It is grounded in the principle that rational argumentation relies on objective information to support a claim (Mohammad et al., 2016). The model acts as an analytical agent, extracting verifiable facts such as statistical data, scientific research findings, or legal precedents. It then evaluates whether this evidence logically supports or contradicts the target. This path is crucial for distinguishing between opinions based on truth claims and those based on mere assertion, thereby enhancing the logical rigor of the stance detection process.

The third reasoning path is **Expert Opinion**. This path models the social influence of authority and expertise, a cornerstone of persuasive communication. In contentious or complex domains, individuals often defer to the judgment of trusted professionals or institutions (e.g., scientists, medical doctors, legal scholars) (Wang et al., 2024a). The model identifies explicit or implicit references to such authoritative sources within the text. If the referenced expert's position aligns with the target (e.g., a scientist stating that climate change is anthropogenic), the stance is inferred as "favor". This path captures the heuristic of "appeal to authority," a common and powerful mechanism in public discourse.

The fourth reasoning path is **Public Opinion**. This path reflects the powerful role of social conformity and normative influence in shaping individual stances. Humans are social beings, and their opinions are often influenced by perceived societal consensus, social media trends, or mainstream media narratives (Wang et al., 2024a). The model identifies phrases that indicate collective sentiment (e.g., "most people think", "everyone is talking about", "public opinion is shifting"). If the perceived public sentiment supports the target, the stance is classified as "favor". This path accounts for the bandwagon effect and the desire to align with the majority.

The selection of these four paths is not arbitrary but is grounded in a comprehensive analysis of human reasoning and empirical validation. They represent a spectrum of influence: from internal affective states (Sentiment) to external rational evidence (Factual), from institutional authority (Expert) to social consensus (Public). This holistic framework ensures that MPRF can capture the multifaceted nature of real-world arguments. Alternative paths, such as *Metaphorical Analysis* (Allaway and McKeown, 2020) and *Temporal Reasoning* (Allaway and McKeown, 2020), were explored but found to be less effective due to ambiguity and sparsity in short texts, respectively. The final four paths thus provide a balanced, theoretically sound, and empirically validated foundation for multi-perspective stance analysis.

3.3 Evaluating Reasoning Paths

Once multiple reasoning paths have been generated, each path is evaluated by a large language model (LLM) based on three main criteria: *relevance*, *evidence strength*, and *logical consistency*. These criteria are inspired by established frameworks in argumentation quality assessment (Hasan and Ng, 2014) and are defined as follows:

- **Relevance Score** r_i**:** Measures how closely the reasoning aligns with the target stance. A higher score indicates stronger alignment.

- **Evidence Strength** e_i**:** Assesses the reliability and robustness of the evidence supporting the reasoning (e.g., factual references to studies or statistics).

- **Logical Consistency** l_i**:** Evaluates whether the reasoning steps follow a coherent and logically sound structure without contradictions.

To ensure consistent and objective scoring across different paths, we designed standardized prompts (see Appendix B) that guide the LLM through structured evaluation steps for each criterion. While human annotation could provide additional validation, our experiments demonstrate a strong correlation between LLM-assigned scores and downstream performance, indicating the practical effectiveness of this automated approach.

After evaluating each reasoning path using these criteria, a final score s_i is assigned to each path by combining the relevance, evidence strength, and logical consistency scores:

$$s_i = \alpha \cdot r_i + \beta \cdot e_i + \gamma \cdot l_i \tag{1}$$

where $i \in \{0, 1, 2, 3\}$ represents the four reasoning paths being evaluated, and α, β, and γ are weights determined through experiments on weighting configurations (Table 5) . The final score s_i reflects the overall quality and contribution of the path to the stance detection task.

3.4 Optimizing Reasoning Paths

After generating and evaluating reasoning paths, we focus on optimizing those that receive low scores in relevance, evidence strength, or logical consistency. Specifically, any path i that score below 5 in any of the evaluation criteria is considered a low-scoring path.

The optimization process involves re-prompting the LLM to refine the reasoning path using a dedicated prompt template (Appendix B), which guides the model to improve specific weak areas—such as enhancing relevance to the target, strengthening supporting evidence, or correcting logical inconsistencies:

$$i' = \mathrm{M}(i, \mathrm{prompt}_{\mathrm{refine}}) \quad \text{if } \min(r_i, e_i, l_i) < 5 \tag{2}$$

Once refined, the updated path i' is re-evaluated using the same scoring criteria:

$$s'_i = \alpha \cdot r'_i + \beta \cdot e'_i + \gamma \cdot l'_i \tag{3}$$

where r'_i, e'_i, l'_i denote the improved relevance, evidence strength, and logical consistency of the optimized path.

This iterative refinement ensures that only high-quality reasoning paths contribute to the final stance prediction, significantly improving both interpretability and accuracy.

3.5 Fusing Reasoning Paths

In step 4, for each reasoning path i, we calculate the weighted score for each label $t \in \{\text{favor}, \text{against}, \text{neutral}\}$. This is done by computing the weighted sum of the individual scores s'_i for each path i assigned to label t, where w_i is the weight of each path based on its contribution to the reasoning process:

$$S_t = \sum_{i \in P_t} w_i \cdot s'_i \tag{4}$$

Here, P_t represents the set of paths assigned to label t, and S_t is the aggregated weighted score for label t.

Next, we calculate the weighted scores for each label (favor, against, neutral) based on the re-evaluated scores for all the reasoning paths: S_{favor}, S_{against} and S_{neutral}.

After the weighted scores for each label are computed, the label with the highest score is selected as the final stance prediction. For example, if $S_{\text{favor}} > S_{\text{against}}$ and $S_{\text{favor}} > S_{\text{neutral}}$, then the final stance is favor. The same approach is applied for the other labels.

In the case where multiple labels have equal scores, the final stance label is determined by a majority vote. Specifically, the label associated with the most reasoning paths is chosen. Even when the scores are equal, the label with more supporting paths generally indicates stronger evidence, making it more likely to represent the true final stance. This voting mechanism resolves tie situations and enhances the robustness of the model by ensuring that the most evidence-backed label is selected.

3.6 Outputting Stance and Reasoning Paths

After determining the final stance prediction in Step 4, we now output both the selected stance label and the corresponding reasoning paths.The reasoning paths that contributed to the selected stance label, illustrating the logical progression and supporting evidence behind the decision. Specifically, the final output consists of:

Final Stance Label: The label selected in Step 4, which can be one of the following: favor, against, or neutral.

Reasoning Paths: The set of reasoning paths associated with the selected stance label, showing how the decision was made. These paths are the ones that contributed to the weighted score for the selected label.

4 Experiments

4.1 Datasets

We conduct experiments on the **VAST**, **SEM16**, and **PStance** datasets to evaluate our proposed method. The **VAST** dataset (Allaway and McKeown, 2020) includes a wide range of targets, each instance comprising a sentence r, a target t, and a stance label y (classified as 'Pro', 'Con' or 'Neutral') towards t.

The **SEM16** dataset (Mohammad et al., 2016) contains six predefined targets, including Donald Trump (DT), Hillary Clinton (HC), Feminist Movement (FM), Legalization of Abortion (LA), Atheism (A), and Climate Change (CC). Each instance is categorized as Favor, Against, or Neutral.

The **PStance** dataset (Li et al., 2021) focuses on the stance of individuals towards three prominent political figures in the United States: Donald Trump (trump), Joe Biden (biden), and Bernie Sanders (sanders). This large-scale dataset includes only two stance labels: favor or against. Detailed statistics of datasets can be found in Appendix A.

4.2 Evaluation Metrics

For the **VAST** dataset, following Allaway and McKeown (2020), we calculate the Macro-averaged F1 score across the *Pro*, *Con*, and *Neutral* labels to evaluate the performance of the models on the test set. For the **SEM16** and **PStance** datasets, we report the F_{avg}, which is the average of the F1 scores for the *Favor* and *Against* classes, in line with Mohammad et al. (2016); Li et al. (2021). We compute F_{avg} for each target. F_{macro} is calculated by averaging the F_{avg} across all targets.

4.3 Baselines

We compare our proposed Multi-Path Reasoning Framework (MPRF) against a comprehensive set of state-of-the-art baselines, categorized into statistics-based models, BERT-based models, and LLM-based models.

Statistics-based models rely on traditional machine learning architectures. We include **BiLSTM** (Schuster and Paliwal, 1997), a bidirectional LSTM without target information; **CNN** (Kim, 2014), a convolutional network also without target modeling; **TAN** (Du et al., 2017), an attention-based LSTM for target-specific features; **BiCond** (Augenstein et al., 2016), which jointly encodes text and target with bidirectional LSTMs; **CrossNet** (Xu et al., 2018), a BiLSTM enhanced with self-attention; **GCAE** (Xue and Li, 2018), a CNN using a gating mechanism to filter irrelevant information; and **PGCNN** (Huang and Carley, 2018), a gated CNN that generates target-sensitive filters.

BERT-based models are built upon the BERT architecture and are typically fine-tuned. We include **BERT** (Devlin et al., 2019), the standard model; **BERTweet** (Nguyen et al., 2020), pre-trained on tweets; **PT-HCL** (Liang et al., 2022a), which uses contrastive learning for zero-shot tasks; **JointCL** (Liang et al., 2022b), a joint contrastive learning framework; **WS-BERT** (He et al., 2022), which infuses Wikipedia knowledge; **CKI** (Yan et al., 2024), a collaborative knowledge infusion

approach; **TATA** (Hanley and Durumeric, 2023), using topic-agnostic and topic-aware embeddings; **KPatch** (Lin et al., 2024), introducing a "knowledge patch" for zero-shot detection; **CNet-Ad** (Zhang et al., 2024c), a commonsense-based adversarial learning framework; and **EZSD-CP** (Yao et al., 2024), a model based on a gated MLP and prompt learning.

LLM-based models leverage the reasoning capabilities of LLMs, often through prompting. We include **KEprompt** (Huang et al., 2023), using knowledge-enhanced prompt-tuning; **KASD** (Li et al., 2023), integrating Wikipedia knowledge with retrieval; **COLA** (Lan et al., 2024), a collaborative role-infusion framework; **Stanceformer** (Garg and Caragea, 2024), a target-aware transformer; **DEEM** (Wang et al., 2024a), using dynamic expert modeling; **GPT-3.5** and **GPT-3.5+COT** (Lan et al., 2024), the base and CoT-prompted versions; **GPT-EDDA** (Ding et al., 2024b), an encoder-decoder data augmentation framework; **Llama-2-7b-chat** and **Llama-2-13b-chat** (Garg and Caragea, 2024), the 7B and 13B chat models; **Manual-CoT** (Wei et al., 2022)and **Auto-CoT** (Zhang et al., 2023c), CoT methods with manual or automatic demonstrations; **ExpertPrompt** (Xu et al., 2023), instructing LLMs to act as experts; **SPP** (Wang et al., 2024b), a solo performance prompting method; and **LC-CoT** (Zhang et al., 2023a), a logically consistent CoT approach.

Detailed descriptions and performance results for these models across different datasets are provided in Appendix C.

4.4 Implementation Details

In our study, we utilize four large language models : **GPT-3.5**(Brown et al., 2020), **Qwen2.5-7B-Instruct**(Qwen Team, 2024) **Llama-3.3-70B-Instruct**(Llama Team, 2024), and **Mistral-7B-Instruct-v0.3**(Jiang et al., 2023). The experiments were conducted on a single NVIDIA A800 GPU with 80GB of RAM, utilizing bfloat16 precision to optimize memory usage and computational efficiency. To ensure the reproducibility of the LLMs' responses, we set the temperature parameter to 0 during inference. This configuration helps maintain consistent outputs across multiple runs. The results reported in our experiments are averaged over 5 repeated runs to ensure statistical reliability and mitigate the impact of any variance in model performance.

5 Results and Discussion

This section addresses the following research questions (RQs) based on our experimental results: **RQ1**: How does the performance of our MPRF compare to state-of-the-art stance detection models on PStance , SEM16 and VAST datasets? **RQ2**: Is each component of the MPRF effective and contributory to overall performance? **RQ3**: How do the weighting configurations of reasoning path scores (s_i') and path weights (w_i) affect the performance of MPRF? **RQ4**: How does the performance of MPRF vary across different models ?

Methods	DT	HC	FM	LA	A	CC
Statistics-based Models						
BiCond	30.5	32.7	40.6	34.4	31.0	15.0
CrossNet	35.6	38.3	41.7	38.5	39.7	22.8
BERT-based Models						
BERT	40.1	49.6	41.9	44.8	55.2	37.3
PT-HCL	50.1	54.5	54.6	50.9	56.5	38.9
TGA-Net	40.7	49.3	46.6	45.2	52.7	36.6
Joint-CL	50.5	54.8	53.8	49.5	54.5	39.7
KPatch	41.1	49.7	43.9	43.8	39.9	31.9
TATA	63.8	65.4	66.9	62.9	52.1	41.6
EZSD-CP	68.8	76.3	62.2	64.4	54.4	37.3
CNet-Ad	47.8	59.2	50.7	54.9	57.4	50.7
TarBK-BERT	50.8	55.1	53.8	48.7	56.2	39.5
LLM-based Models						
KASD-ChatGPT	-	80.3	70.4	62.7	-	-
COLA	68.5	81.7	63.4	71.0	70.8	65.5
GPT-3.5	62.5	68.7	44.7	51.5	9.1	31.1
GPT-3.5+COT	63.3	70.9	47.7	53.4	13.3	34.0
GPT-EDDA	69.5	80.1	69.2	62.7	67.2	68.5
LCDA	70.0	79.8	70.0	69.4	-	-
LC-CoT	71.7	82.9	70.4	63.2	-	-
MPRF (Ours)						
GPT-3.5	80.3	83.5	83.2	84.1	80.4	82.9
Qwen2.5-7B-Instruct	**84.4***	83.1*	84.5*	82.5	83.9*	82.3*
LLaMA-3.3-70B	81.5	83.4	83.4	**85.3***	**84.3***	**83.6***
Mistral-7B-Instruct-v0.3	82.6	**84.5***	**87.7***	83.2	82.7	83.4

Table 1: Zero-shot stance detection results on the SEM16 dataset. The best scores are in bold.Results with * denote that MPRF significantly outperforms baselines with the p-value < 0.05.

Performance Comparison with State-of-the-Art Models As shown in Tables 1, 2, and 3, MPRF significantly outperforms state-of-the-art models across all three datasets—SEM16, VAST, and PStance—demonstrating both effectiveness and innovation in stance detection.

On SEM16, MPRF achieves the highest scores across multiple targets (e.g., FM: 87.7%, HC: 84.5%), surpassing strong LLM-based baselines such as GPT-EDDA and KASD by a large margin. In particular, our method shows robust generalization to challenging targets like "Atheism" and "Climate Change", where traditional models often struggle.

On VAST, MPRF achieves an overall F1 score of 83.5% with Mistral-7B-Instruct-v0.3, outperforming existing methods by more than 3%. Notably, it performs exceptionally well in zero-shot settings (85.4%), highlighting its ability to generalize without target-specific training data.

On PStance, MPRF sets a new state of the art with an F_{macro} of 88.41%, surpassing DEEM by 3.26%. The model also achieves the best individual performance on Biden (89.36%) and Trump (89.33%), demonstrating strong adaptability to real-world political discourse. These results affirm that MPRF's structured multi-path reasoning framework brings substantial improvements over conventional single-path or static prompting approaches, offering both enhanced accuracy and interpretability.

Model	Zero-Shot	Few-Shot	Overall
BiCond	42.8	40.0	41.5
CrossNet	43.4	47.4	45.5
TGA-Net	66.6	66.3	66.5
BERT-GCN	68.6	69.7	69.2
CKE-Net	70.2	70.1	70.1
GDA-CL	70.5	-	-
PT-HCL	71.6	-	-
WS-BERT	75.3	73.6	74.5
CNet-Ad	73.2	71.8	72.6
TarBK-BERT	73.6	-	-
Joint-CL	72.3	71.5	-
EZSD-CP	73.6	-	-
StSQA	68.9	-	-
COLA	73.4	-	-
TATA	77.1	74.1	76.3
GPT-3.5-Turbo	65.0	-	-
GPT-3.5-Turbo-CoT	66.4	-	-
KASD-BERT	76.8	-	-
LKI-BART	79.6	-	-
CKI	81.9	79.6	80.7
EDDA-LLaMA	70.3	-	-
LC-CoT	72.5	-	-
LCDA	80.3	-	-
MPRF (Ours)			
GPT-3.5	81.1	82.3	82.1
Qwen2.5-7B-Instruct	**85.4***	82.0	83.2
LLaMA-3.3-70B	80.6	82.1	81.2
Mistral-7B-Instruct-v0.3	84.2*	**83.1***	**83.5***

Table 2: Stance detection performance (%) on VAST. The best scores are in bold.Results with * denote that MPRF significantly outperforms baselines with the p-value < 0.05.

Contribution of Each Component of MPRF To evaluate the effectiveness of each component, we conducted ablation studies on SEM16 (Table 4), with additional results for PStance and VAST in Appendix D. Removing any component leads to performance drops, confirming their importance.

Sentiment Path significantly influences targets involving emotional expression. Its removal causes a 4.6% drop on FM (from 87.7% to 83.1%) on SEM16 and a 1.9% drop on Trump (from 89.33% to 87.4%) on PStance.

Method	Trump	Biden	Sanders	F_{macro}
BiLSTM	76.92	77.95	69.75	74.87
CNN	76.80	77.22	71.40	75.14
TAN	77.10	77.64	71.60	75.45
BiCond	77.15	77.69	71.24	75.36
PGCNN	76.87	76.60	72.13	75.20
GCAE	78.96	77.95	71.82	76.24
BERT	78.28	78.70	72.45	76.48
BERTweet	82.48	81.02	78.09	80.53
WS-BERT	84.97	82.86	79.97	82.60
TarBK-BERT	65.80	75.49	70.45	70.58
GPT-3.5	79.80	79.65	77.77	79.07
Llama-2-7b-chat-ft	72.00	67.96	65.57	68.51
Llama-2-13b-chat-ft	76.62	71.88	68.44	72.31
Stanceformer	85.35	83.96	80.57	83.30
CKI	86.2	84.1	80.5	83.6
Manual-CoT	85.4	83.8	80.9	83.37
StSQA	85.7	82.8	80.8	83.1
Auto-CoT	84.1	82.8	80.6	82.5
ExpertPrompt	84.7	84.7	81.2	83.53
SPP	85.1	84.6	81.5	83.73
DEEM	86.4	86.1	82.1	84.87
COLA	86.6	84.0	79.7	83.1
KASD-ChatGPT	85.06	84.59	79.96	83.2
MPRF (Ours)				
GPT-3.5	88.20	88.60	85.10	87.60
Qwen2.5-7B-Instruct	89.50*	88.15*	**86.75***	88.13
LLaMA-3.3-70B	**89.63***	88.26	85.35	87.75
Mistral-7B-Instruct-v0.3	89.33	**89.36***	86.53	**88.41***

Table 3: Comparison of different models on the PStance . The best scores are in bold.Results with * denote that MPRF significantly outperforms baselines with the p-value < 0.05.

Fact-Based Reasoning is crucial for evidence-based stance formation. Without it, FM drops by 3.7% on SEM16 and Trump drops by 1.6% on PStance.

Expert Opinion Path enhances predictions through authoritative viewpoints. Removing it leads to a 3.4% drop on FM and a 1.2% drop on Trump.

Public Opinion Path contributes moderate but consistent improvements, especially in social context modeling. Its removal results in a 3.3% drop on FM and a 1.3% drop on Trump.

Optimization plays a key role in refining low-quality paths. Without optimization, FM drops sharply by 7.1%, indicating its critical impact on reasoning quality.

Weighted Fusion has the most significant impact among all components. Removing it causes an 8.6% drop on FM and a 3.6% drop on Trump, highlighting its essential role in integrating diverse perspectives effectively.

Model	DT	HC	FM	LA	A	CC
MPRF	**82.6**	**84.5**	**87.7**	**83.2**	**82.7**	**83.4**
w/o Sentiment Path	80.3	81.2	83.1	81.4	81.8	81.9
w/o Fact-Based Path	81.1	82.4	84.0	82.6	82.3	82.0
w/o Expert Opinion Path	81.5	82.8	84.3	82.9	83.1	82.6
w/o Public Opinion Path	81.7	83.1	84.4	83.0	83.3	82.9
w/o Optimization	79.7	80.4	80.6	80.3	80.5	80.0
w/o Weighted Fusion	78.9	79.8	79.1	79.6	79.8	78.5

Table 4: Ablation Study Results on SEM16 (%) using Mistral-7B-Instruct. The best scores are in bold.

Impact of Weighting Configurations on MPRF Performance Table 5 shows the impact of different weighting configurations on MPRF's performance. The optimal configuration ($\alpha = 0.4, \beta = 0.3, \gamma = 0.3$) yields the highest F1 scores, including 87.7% for FM and 84.5% for HC, outperforming other configurations. This demonstrates that a balanced weighting prioritizes relevance while maintaining adequate consideration of evidence strength and logical consistency.

Other configurations, such as the equal weighting ($\alpha = 0.33, \beta = 0.33, \gamma = 0.33$), result in significant performance drops, with FM and HC scores dropping to 83.1% and 81.2%, respectively. Skewed configurations that emphasize a single criterion—like relevance ($\alpha = 0.5, \beta = 0.25, \gamma = 0.25$), evidence ($\alpha = 0.25, \beta = 0.5, \gamma = 0.25$), or logical consistency ($\alpha = 0.25, \beta = 0.25, \gamma = 0.5$) also perform poorly. These configurations fail to balance the complementary contributions of all components, leading to suboptimal results.

Weighting Configurations	DT	HC	FM	LA	A	CC
MPRF	**82.6**	**84.5**	**87.7**	**83.2**	**82.7**	**83.4**
Balanced Weights	80.3	81.2	83.1	81.4	81.8	81.9
Enhanced Relevance	81.4	82.2	81.0	80.4	81.1	81.6
Enhanced Evidence Strength	81.8	83.5	82.1	81.3	80.0	82.0
Enhanced Logical Consistency	81.4	82.7	82.9	83.0	80.6	81.8

Table 5: Effect of Weighting Configurations on SEM16 (%) using Mistral-7B-Instruct. The best scores are in bold.

Table 6 illustrates the impact of different weighting configurations on the SEM16 dataset. The MPRF ($w_i \propto s_i'$) configuration, which dynamically adjusts path weights based on their scores (s_i'), achieves the best performance, with F1 scores of 87.7% for FM and 84.5% for HC, outperforming all other configurations.

In contrast, the Equal Weights configuration shows a significant performance drop, with FM and HC scores decreasing to 83.1% and 81.2%, respectively. Configurations assigning high weights (0.8) to specific paths, such as sentiment or fact-based reasoning, also fail to surpass the MPRF ($w_i \propto s_i$) configuration.

The superiority of dynamic weighting highlights the importance of prioritizing high-quality reasoning paths. Equal weighting dilutes the contributions of critical paths, while overemphasizing a single path reduces the complementary contributions from other paths.

In conclusion, dynamically adjusting weights based on path scores is crucial for maximizing performance, while equal or skewed weighting results in significant performance decline, emphasizing the need for balanced contributions from multiple reasoning paths.

Weighting Configurations	DT	HC	FM	LA	A	CC
MPRF ($w_i \propto s_i'$)	**82.6**	**84.5**	**87.7**	**83.2**	**82.7**	**83.4**
Equal Weights	80.3	81.2	83.1	81.4	81.8	81.9
High for Sentiment Path	81.4	82.8	84.3	81.9	81.6	82.1
High for Fact-Based Path	81.8	83.7	85.1	82.6	82.0	82.5
High for Expert Path	80.8	82.3	83.5	82.0	81.7	82.2
High for Public Path	80.9	82.5	83.8	81.8	81.5	81.9

Table 6: Comparison of Weighting Configurations (%) on SEM16 using Mistral-7B-Instruct. The best scores are in bold.

Performance of MPRF Across Different Models Table 1, Table 2, and Table 3 summarize the performance of MPRF across four distinct large language models on the SEM16, VAST, and PStance datasets. These results demonstrate that MPRF consistently improves stance detection performance across diverse model architectures, suggesting that its effectiveness stems from the framework design rather than specific model capabilities.

On SEM16, for example, all four models achieve strong results under MPRF, with Mistral-7B-Instruct-v0.3 performing best on FM (87.7%) and HC (84.5%), and Qwen2.5-7B-Instruct leading on DT (84.4%). The relatively smaller performance gaps between models suggest that MPRF mitigates inherent model-specific biases or limitations by leveraging structured reasoning paths and weighted fusion. Similarly, in zero-shot and few-shot settings on VAST and PStance, MPRF enables consistent improvements across models.

The key takeaway is that MPRF's performance gains are not solely attributable to the choice of LLM, but rather to its ability to generate, evaluate, and integrate multiple reasoning paths effectively. This consistency across models confirms that MPRF provides a robust and model-agnostic enhancement to stance detection frameworks.

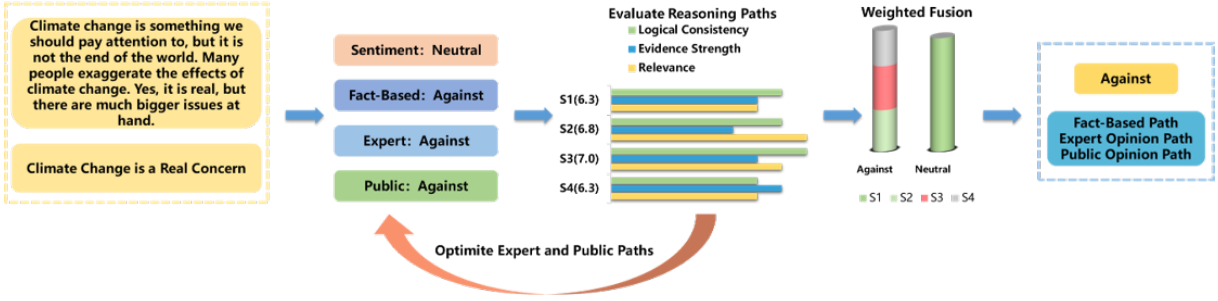


Figure 2: Stance Detection Process: Multi-Path Analysis of the Statement on Climate Change

Case Study: Stance Detection on "Climate Change is a Real Concern" To demonstrate the interpretability of our Multi-Path Reasoning Framework (MPRF), we present a detailed case study in Figure 2: *"Climate change is something we should pay attention to, but it's not the end of the world. Many people exaggerate the effects of climate change. Yes, it's real, but there are much bigger issues at hand."* Our goal is to detect the stance towards the target *"Climate Change is a Real Concern"*.

MPRF analyzes this statement through four distinct, human-understandable reasoning paths, each representing a different dimension of opinion formation. First, the **Sentiment Analysis** path identifies the overall tone as neutral. The text acknowledges the reality of climate change ("it's real") but tempers this with phrases like "not the end of the world" and "many people exaggerate," resulting in a balanced emotional expression and a stance prediction of *Neutral*. Second, the **Factual Reasoning** path focuses on the evidence presented. The claim that "many people exaggerate the effects" is interpreted as a direct challenge to the severity of the scientific consensus on climate change, leading to a stance prediction of *Against*. Third, the **Expert Opinion** path infers that the statement contradicts the overwhelming consensus of climate scientists. By downplaying the urgency and scale of the issue, the argument implicitly opposes the authoritative viewpoint, resulting in a stance of *Against*. Fourth, the **Public Opinion** path interprets the phrase "many people exaggerate" as a reflection of a perceived public sentiment that overstates the problem, thus classifying the stance as *Against*.

Crucially, MPRF does not treat these paths equally. Each path is rigorously evaluated by an LLM on three criteria: relevance, evidence strength, and logical consistency, and assigned a quantitative score. For instance, the Fact-Based path received a high relevance score (8) due to its direct engagement with the core issue, though its evidence strength was moderate (5). The Expert Opinion path scored highly in logical consistency (8), reflecting a coherent argument structure. These individual scores (s_i') are then used in a weighted fusion mechanism. The paths predicting *Against* (Fact-Based, Expert Opinion, Public Opinion) collectively contribute a significantly higher aggregated weighted score (S_{against}) than the single *Neutral* path. This transparent, score-driven aggregation process results in the final prediction of *Against*.

6 Conclusion

In this paper, we proposed the Multi-Path Reasoning Framework (MPRF), a novel framework for stance detection that enhances accuracy, robustness, and interpretability by generating, evaluating, and optimizing multiple reasoning paths. Through experiments on three datasets, SEM16, VAST, and PStance, MPRF consistently demonstrated strong performance across various models. The results highlight the adaptability and effectiveness of MPRF in both zero-shot and few-shot settings, achieving state-of-the-art performance and providing interpretable reasoning chains.

Despite these achievements, there are still some limitations to our approach. The reliance on manually designed prompts for path evaluation and scoring may limit scalability to more complex tasks or datasets. In future work, we aim to explore automated prompt optimization and more efficient reasoning strategies to further enhance MPRF's scalability and performance in real-world applications.

Limitations

While the Multi-Path Reasoning Framework (MPRF) demonstrates strong performance and adaptability across various datasets and models, there are several limitations to address. First, the reliance on manually designed prompts for evaluating and scoring reasoning paths may limit scalability to more complex tasks or domains where predefined prompts may not generalize well. Second, generating and optimizing multiple reasoning paths introduces additional computational overhead, which could hinder its deployment in large-scale or time-sensitive applications. Lastly, the interpretability of MPRF, while improved compared to traditional black-box models, still requires further exploration to ensure that reasoning paths are both human-readable and aligned with domain-specific requirements.

Ethical Considerations

The Multi-Path Reasoning Framework(MPRF) aims to improve stance detection by enhancing transparency and interpretability in its decision-making process. However, the use of large language models (LLMs) in MPRF raises potential ethical concerns. First, LLMs are trained on vast amounts of internet data, which may include biased or harmful content, potentially influencing the reasoning paths generated by MPRF. Second, the interpretability of the reasoning paths may lead to unintended misuse, such as generating misleading or manipulative arguments. Finally, while MPRF strives for fairness, it inherits biases present in the underlying models and datasets, which could result in unfair or inaccurate stance detection in certain contexts. To mitigate these risks, we have carefully evaluated MPRF's outputs and employed diverse datasets to reduce potential biases. Future work will focus on improving fairness, bias detection, and ethical safeguards to ensure responsible application of MPRF in real-world scenarios.

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A Dataset Statistics

	Train	Dev	Test
# Examples	13477	2062	3006
# Unique Comments	1845	682	786
# Zero-shot Topics	4003	383	600
# Few-shot Topics	638	114	159

Table 7: Statistics of **VAST** dataset.

Target	Favor	Against	Neutral
DT	148	299	260
HC	163	565	256
FM	268	511	170
LA	167	544	222
A	124	464	145
CC	335	26	203

Table 8: Statistics of **SEM16** dataset.

		Trump	Biden	Sanders
Train	**Favor**	2,937	2,552	2,858
	Against	3,425	3,254	2,198
Val	**Favor**	365	328	350
	Against	430	417	284
Test	**Favor**	361	337	343
	Against	435	408	292
Total		7,953	7,296	6,325

Table 9: Label distribution across different targets for P-Stance.

B Prompts

We selected these four reasoning paths—sentiment analysis, factual reasoning, expert opinions, and public opinion—based on their semantic distinctiveness and empirical effectiveness in capturing key dimensions of stance reasoning. Each path addresses a unique aspect that commonly influences human judgment:

- **Sentiment Analysis Path**: Captures emotional tone (e.g., "happy", "angry") to infer stance. Emotional language often strongly correlates with stance expression.

- **Factual Reasoning Path**: Analyzes factual evidence (e.g., statistics, legal references) to determine alignment with the target stance. Stance decisions frequently hinge on objective claims or data.

- **Expert Opinion Path**: Evaluates references to authoritative sources (e.g., scientists, legal scholars). Authority-driven arguments are particularly influential in contentious or technical topics.

- **Public Opinion Path**: Reflects collective sentiment derived from social media, surveys, or news coverage. Public consensus can significantly shape individual stances through social influence.

Prompt Template for Sentiment Path:

> Your task is to classify the stance of the comment on the topic as “favor”, “against”, or “neutral”.
> Topic: Target
> Comment: Text
> Stance:
> Explain:
> 1. Analyze the emotional tone of the comment. Identify key emotional words such as “happy”, “sad”, “angry”, “disappointed”, “excited”, etc.
> 2. Based on the sentiment, determine if the comment expresses a positive, negative, or neutral emotion.
> 3. If the comment expresses a positive sentiment, classify the stance as “favor”; if negative, classify the stance as “against”; if neutral, classify the stance as “neutral”.

Prompt Template for Fact-Based Reasoning Path:

> Your task is to classify the stance of the comment on the topic as “favor”, “against”, or “neutral”.
> Topic: Target
> Comment: Text
> Stance:

Explain:
1. Extract factual information from the comment, such as data, studies, or legal references related to the target.
2. Determine if the facts support or oppose the target stance.
3. If the facts present a positive outcome or support legalization, classify the stance as "favor"; if they present negative impacts or oppose it, classify the stance as "against"; if the facts are neutral or inconclusive, classify the stance as "neutral".

Prompt Template for Expert Opinion Path:

Your task is to classify the stance of the comment on the topic as "favor", "against", or "neutral".
Topic: Target
Comment: Text
Stance:
Explain:
1. Identify expert opinions or references within the comment.
2. Determine if the expert opinions support or oppose the target stance.
3. If the expert opinions support the target, classify the stance as "favor"; if they oppose it, classify the stance as "against"; if the expert opinions are neutral or inconclusive, classify the stance as "neutral".

Prompt Template for Public Opinion Path:

Your task is to classify the stance of the comment on the topic as "favor", "against", or "neutral".
Topic: Target
Comment: Text
Stance:
Explain:
1. Extract public opinion from the comment, such as social media reactions, public surveys, or common sentiments.
2. Determine if the public opinion supports or opposes the target.
3. If the public opinion supports target, classify the stance as "favor"; if it opposes, classify the stance as "against"; if the public opinion is neutral, classify the stance as "neutral".

Prompt for Relevance Score:

Please evaluate the relevance of the following reasoning path: reasoning path, in relation to the target Target, given the comment: Text.
How relevant is the reasoning path to the target Target? Please provide a relevance score from 1 to 10, where 1 represents low relevance and 10 represents high relevance.

Prompt for Evidence Strength Score:

Evaluate the strength of the evidence in the following reasoning path :reasoning path for Target. Does the evidence strongly support the stance of target? Please provide an evidence strength score from 1 to 10, where 1 indicates weak or unreliable evidence and 10 indicates strong and reliable evidence.

Prompt for Logical Consistency Score:

Evaluate the logical consistency of the following reasoning path:reasoning path for Target. Are the reasoning steps coherent and logically connected? Please provide a logical consistency score from 1 to 10, where 1 indicates poor consistency and 10 indicates excellent consistency.

Prompt Template for Optimization of Low-Scoring Reasoning Paths:

Your task is to review and improve the following reasoning path for Target. This path has been evaluated and received low scores in one or more of the following areas: Relevance Score r_i, Evidence Strength Score e_i, or Logical Consistency Score l_i. The scores below 5 indicate weak areas that need optimization. The initial reasoning path is:

reasoning path

Please focus on improving the areas where the scores are low. Specifically, address the following:

1. Low Relevance Score (r_i): - Review the reasoning steps that have been marked as irrelevant or weakly connected to the target stance. Adjust these steps to ensure they are directly relevant to the target. If necessary, provide stronger connections to the target stance.

2. Low Evidence Strength Score (e_i): - Identify reasoning steps that lack strong supporting evidence or facts. Propose more robust evidence, studies, or expert opinions to support these steps and strengthen the overall argument.

3. Low Logical Consistency Score (l_i): - Examine any steps that show logical inconsistencies or errors. Identify these inconsistencies and suggest improvements to fix any logical flaws, ensuring the reasoning follows a consistent and sound structure.

After revising the weak areas, provide the updated reasoning path with all improvements incorporated.

Revised reasoning path:

C Baselines

In this appendix, we provide detailed descriptions and performance results for the baseline models compared to MPRF across the SEM16, PStance, and VAST datasets.

C.1 Statistics-based Models

We include **BiLSTM**(Schuster and Paliwal, 1997), which takes tweets as inputs without considering the target information; **CNN**(Kim, 2014), similar to BiLSTM, the vanilla CNN only takes tweets as inputs and does not consider the target information; **TAN**(Du et al., 2017) an attention-based LSTM model that extracts target specific features ; **BiCond** (Augenstein et al., 2016), which uses bidirectional LSTM to encode both the text and the target; **CrossNet** (Xu et al., 2018), which enhances BiLSTM with self-attention mechanisms to improve focus on relevant text segments; **GCAE**(Xue and Li, 2018) , a CNN model that utilizes a gating mechanism to block targetunrelated information; **PGCNN**(Huang and Carley, 2018), which is based on gated convolutional networks and encodes target information by generating target-sensitive filters.

C.2 BERT-based Models

We benchmark against **BERT** (Devlin et al., 2019), a transformer model fine-tuned for stance detection, **PT-HCL** (Liang et al., 2022a), which incorporates contrastive learning within a BERT framework targeting zero-shot and cross-target tasks. **BERT-joint-ft**(Liu et al., 2021) and **TGA-Net-ft**(Liu et al., 2021), in which BERT has been fine-tuned during the training process. **BERTweet**(Nguyen et al., 2020), **CKE-Net**(Liu et al., 2021), **GDA-CL**(Li and Yuan, 2022), **WS-BERT**(He et al., 2022), **BS-RGCN**(Luo et al., 2022), **BERT-joint**(Allaway and McKeown, 2020), **TGA Net** (Allaway and McKeown, 2020), **JointCL** (Liang et al., 2022b), **KPatch**(Lin et al., 2024), **TATA**(Hanley and Durumeric, 2023), **EDDA-LLaMA**(Ding et al., 2024b); **EZSD-CP**(Yao et al., 2024), a stance detection model underpinned by a gated multilayer perceptron and a prompt learning strategy ;**CKI** (Yan et al., 2024), a collaborative knowledge infusion approach , BERT-based models showing strong performance on datasets; **CNet-Ad**(Zhang et al., 2024c), a commonsense based adversarial learning framework that comprises a commonsense graph encoder and a feature separation adversarial network.

C.3 LLM-based Models

We compare with **KEprompt** (Huang et al., 2023), which leverages knowledge-enhanced prompt tuning; **KASD** (Li et al., 2023), enhancing detection by integrating Wikipedia knowledge with retrieval-enhanced generation; **COLA** (Lan et al., 2024), which employs a collaborative role-infusion framework with multiple LLMs;**Llama-2-7b-chat2** , **Llama-2-13b-chat** and finetuned settings for the Llama models(Garg and Caragea, 2024);**Manual-CoT** (Wei et al., 2022) manually provides the explanations in demonstrations and enhances the chainof-thought reasoning ability of LLMs; **StSQA**(Zhang et al., 2024b) proposes automatic "thought-inducing" and add them to the demonstrations for step-by-step question answering; **Auto-CoT** (Zhang et al., 2023c) automatically selects demonstrations from training data according to semantic diversity; **ExpertPrompt**(Xu et al., 2023) introduces the identity of experts and customizes information descriptions for LLMs before giving responses; **SPP** (Wang et al., 2024b) proposes

solo performance prompting by engaging in multi-turn collaboration with multi-persona during reasoning; **DEEM**(Wang et al., 2024a), dynamic experienced expert modeling for stance detection; **Stanceformer**(Garg and Caragea, 2024), a target-aware transformer model; **GPT-3.5** (Lan et al., 2024), **GPT-3.5+COT** (Lan et al., 2024), **GPT-EDDA**(Ding et al., 2024b), and **LKI-BART**(Zhang et al., 2024d) .**LC-CoT**(Zhang et al., 2023a) employs the structured chain-of-thought approach for stance detection. **LCDA** enhances data quality by maintaining logical coherence (Zhang et al., 2025)

D Ablation Study Results

In this appendix, we present the ablation study results for the VAST and PStance datasets. These tables illustrate the performance of the model with various components removed. The results demonstrate the importance of each component in stance detection.

D.1 Ablation Study Results on VAST

Model	Zero-Shot	Few-Shot	Overall
Mistral-7B-Instruct-v0.3	**84.2**	**83.1**	**83.5**
w/o Sentiment Path	82.7	81.4	81.9
w/o Fact-Based Path	83.2	82.1	82.5
w/o Expert Opinion Path	83.0	82.5	82.7
w/o Public Opinion Path	83.1	81.9	82.5
w/o Optimization	81.6	80.5	80.9
w/o Weighted Fusion	80.8	79.3	79.9

Table 10: Ablation Study Results on VAST (%) using Mistral-7B-Instruct-v0.3. The best scores are in bold.

The results on VAST using Mistral-7B-Instruct-v0.3 are shown in Table 10. The performance decreases when components like optimization and weighted fusion are omitted, highlighting their importance for improving the final stance prediction.

D.2 Ablation Study Results on PStance

Method	Trump	Biden	Sanders	F_{macro}
MPRF	**89.33**	**89.36**	**86.53**	**88.41**
w/o Sentiment Path	87.4	85.9	84.0	85.4
w/o Fact-Based Path	87.7	86.3	84.5	85.8
w/o Expert Opinion Path	88.1	87.0	85.1	86.2
w/o Public Opinion Path	88.0	86.5	84.8	85.8
w/o Optimization	86.5	85.1	83.4	84.6
w/o Weighted Fusion	85.7	84.2	82.8	83.8

Table 11: Ablation Study Results on PStance (%) using Mistral-7B-Instruct-v0.3. The best scores are in bold.

The results on PStance using Mistral-7B-Instruct-v0.3 are shown in Table 11. The table shows the performance drop when certain components, such as the sentiment path, fact-based path, and others, are removed.